TRADE INFRASTRUCTURE LTD
Sales Cowboy | Enterprise Digital & Payment Solutions

INSTITUTIONAL PROPOSAL
Ref: SC-PROP-2026-FINTECH
Date: November 8, 2026
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INSTITUTIONAL PRODUCT & COMPLIANCE ARCHITECTURE

Cross-Border Stablecoin & Multi-Currency Settlement Engine

Multi-Platform Client Apps (Web/PWA, iOS, Android) + Core Banking Engine + Institutional Liquidity & Compliance Hub

1. EXECUTIVE SUMMARY & ARCHITECTURAL APPROACH

Deploying a multi-currency fiat-to-stablecoin payment processor across West Africa involves navigating two major challenges: fragmented inter-bank clearing rails (NGN NIBSS/NUBAN, GHS GhIPSS, XOF BCEAO/GIM-UEMOA) and strict AML/CFT/VASP regulatory regimes. Building proprietary banking rails, independent clearing networks, and separate financial licenses in every jurisdiction from day one requires tens of millions of dollars and multi-year licensing delays.

The institutional approach utilized by modern African cross-border fintechs is a **Licensed BaaS (Banking-as-a-Service) & Regulated Liquidity Proxy Model**:

1. Fiat Rails & Local Clearing: Integrated via regulated tier-1 payment service providers and commercial banks holding local clearing licenses (Korapay, Flutterwave, Paystack, or Yellow Card across Nigeria, Ghana, Kenya, and UEMOA markets).
2. Crypto & Stablecoin Infrastructure: Built on dedicated MPC (Multi-Party Computation) custodial wallets (Fireblocks or BitGo API architecture) connected to institutional OTC desks for automated, zero-slippage USDT/USDC-to-fiat conversion.
3. Card Issuing & Bill Payments: Sourced via licensed card scheme infrastructure partners (Mastercard/Visa enablement bridges like Bridgecard or Maplerad).

2. FUNCTIONAL SYSTEM SCOPE

Module 1: Client Apps & Core Ledger

- **KYC / AML Engine:** Tier 1 to 3 automated verification (BVN/NIN, Ghana Card, Passport, biometric liveness, PEP/Sanctions screening via Smile ID / Dojah).
- **Multi-Currency Wallets:** Dedicated accounts for NGN, GHS, XOF alongside hybrid custodial USDT (TRC20, ERC20, Polygon).
- **On/Off Ramps:** Dedicated Virtual Accounts (DVA) for instant bank deposits, local transfers, and Mobile Money (MTN, Telecel, Orange) payouts.
- **Instant FX Swaps:** Real-time quote engine calculating cross-currency rates with dynamic spread margins (0.5% to 1.5%).
- **Cross-Border P2P & B2B:** Instant corridors settlement bypassing correspondent banking delays.
- **Virtual Cards & VAS:** USD/NGN virtual Mastercard/Visa issuance, airtime, data bundles, and cross-border utilities.

Module 2: Institutional Engine & Treasury

- **Double-Entry Ledger:** ACID-compliant transaction isolation and automated reserve accounting preventing phantom balances.
- **Automated OTC Rebalancing:** Webhook-driven auto-swaps via institutional OTC endpoints to hedge currency volatility upon crypto receipt.
- **MPC Custodial Security:** Distributed key management splitting signing across independent nodes, eliminating single-point theft risk.
- **Automated Liquidity Sweeps:** Threshold-based sweeping into cold storage vaults and high-yield reserve accounts.
- **High-Throughput Microservices:** NestJS microservice mesh engineered for concurrent transaction execution.

Module 3: Compliance & Admin Hub

- **Transaction Monitoring (STR/SAR):** Automated velocity limits, suspicious activity alerts, and AML watchlist checks.
- **On-Chain AML Screening:** Real-time blockchain address screening via Chainalysis / Elliptic API bridge.
- **Treasury Float Dashboard:** Unified balance visibility across partner commercial banks, OTC desks, and on-chain liquidity pools.
- **Dynamic Spread & Fee Matrix:** Real-time configuration of swap spreads, gas markups, card issuance fees, and payout commissions.
- **Regulatory Audit Reporting:** Automated export of daily transaction logs and compliance audit trails for Central Bank inspection.

3. REGULATORY FRAMEWORK & LICENSING ROADMAP

Operating legally across West Africa requires a dual strategy: launching rapidly via **authorized sponsorship licenses (Phase 1)**, while initiating multi-stage corporate filings for proprietary recognition (Phase 2).

Phase 1: Operational Launch (License Sponsorship Model)

• **Legal Setup:** Corporate holding entity paired with local subsidiaries. Compliant AML/CFT policies, User Terms, Cross-Border Disclosures, and NDPR / African Data Privacy frameworks.
• **Operational Sponsorship:** Partnering with accredited institutions holding Nigerian PSSP / Switching and SEC VASP registrations, Bank of Ghana (BoG) PSP Enhanced licenses, and BCEAO-accredited Electronic Money Institutions (EMI).

Phase 2: Proprietary Licensing & Direct Capital Requirements

JURISDICTION	REGULATORY BODY	REQUIRED LICENSE	STATUTORY CAPITAL & OFFICIAL FILING ESTIMATES
Nigeria	Securities & Exchange Commission (SEC)	**VASP License** / ARIP Registration	Minimum capital requirement (~₦1 Billion unimpaired capital) + SEC processing fees (~₦30M to ₦50M).
Nigeria	Central Bank of Nigeria (CBN)	**PSSP / IMTO License**	CBN statutory capital deposit (~$1M or ₦2B depending on tier) + official application fees.
Ghana	Bank of Ghana (BoG)	**PSP Enhanced License**	GHS 2M to GHS 5M capital requirements + technical and security audit vetting.
Global / Offshore	Financial Services Commission (BVI / Mauritius)	**International Money Services / VASP**	$25,000 to $60,000 legal formation, resident directorship, and statutory capital deposits.

(Note: Direct statutory capital reserves are corporate equity deposits held by Central Banks, not software development costs. Operations launch under Phase 1 sponsorship while funding Phase 2).

4. ESTIMATED THIRD-PARTY OPERATIONAL COSTS

Recurring infrastructure, custodial, and compliance integration fees contracted directly with accredited providers:

COMPONENT / SERVICE	RECOMMENDED PROVIDER	ESTIMATED OPERATIONAL / MONTHLY COST
Cloud Infrastructure (PCI-DSS Ready)	AWS / Google Cloud Enterprise Cluster + Redis	**$250.00 to $600.00 / month** (Scales with transaction volume)
Crypto Custody & MPC Vaults	Fireblocks / BitGo (or hybrid self-hosted MPC)	**$1,000.00 to $2,500.00 / month** (plus marginal volume tiers)
KYC / AML Screening Engine	Smile ID / Dojah / Sumsub	**Pay-as-you-go** (~$0.40 to $1.20 per verified user)
Blockchain Screening & Analytics	Chainalysis / Crystal / AMLBot API	**$200.00 to $500.00 / month** base subscription
Virtual Card Issuing Program	Bridgecard / Maplerad / Sudo Africa	**$1,500 to $3,000 setup fee** + interchange revenue-share
SMS OTP & Alert Delivery	Termii / Twilio Global Gateway	**Pay-as-you-go** (~₦5.00 local, ~$0.04 international per SMS)

5. ENGINEERING INVESTMENT & IMPLEMENTATION SCHEDULE

Total Full-Stack Engineering Investment: ₦12,000,000 (Twelve Million Naira) for end-to-end development of the cross-border stablecoin engine, multi-currency banking rails, mobile client apps (iOS/Android), and executive compliance console over 16 weeks.

PHASE	SCOPE & KEY DELIVERABLES	INVESTMENT	TIMELINE
Phase 1: Core Architecture & Liquidity Engine	PostgreSQL double-entry database schema, high-throughput NestJS microservices, USDT MPC custodial wallet engine, and automated OTC liquidity integration.	**₦4,500,000**	6 Weeks
Phase 2: Banking Rails & FX Swap Desk	Local fiat on/off ramps (NGN, GHS, XOF), instant cross-currency FX swap engine, international P2P transfer module, and automated KYC verification gates.	**₦4,000,000**	6 Weeks
Phase 3: Cards, VAS & Production Hardening	Virtual card provisioning, bill payments & VAS, cross-platform mobile apps (iOS & Android via React Native), Admin console, and security penetration audit.	**₦3,500,000**	4 Weeks
Total Full-Stack Enterprise Deployment		**₦12,000,000**	**16 Weeks**

Commercial Milestone Breakdown

MILESTONE STAGE	DELIVERABLE TRIGGER	FEE BREAKDOWN
1. Mobilization Deposit	System architecture, security model, database schema, and core ledger execution.	**35% (₦4,200,000)**
2. Staging & Integration	Completion of fiat/crypto gateways, OTC swap mechanics, and staging deployment.	**35% (₦4,200,000)**
3. Production Handover	Live mainnet and production banking rail testing, mobile app distribution setup, and administrative compliance handover.	**30% (₦3,600,000)**

6. POST-LAUNCH WARRANTY & ENTERPRISE SLA

Following production deployment and mainnet handover, Sales Cowboy provides **sixty (60) days of complimentary tier-1 technical warranty support** covering:

- Real-time ledger audit monitoring, database integrity surveillance, and reconciliation validation.
- Zero-downtime hotfix deployments for critical bug remediation and edge-case transaction resolution.
- Third-party banking gateway and custodial webhook listener resilience and failover optimization.
- Dedicated technical onboarding sessions for executive compliance officers, treasury managers, and support desk staff.

FOR: SALES COWBOY (DEVELOPER)

Lead Fintech Architect & Systems Engineer
Sales Cowboy / Trade Infrastructure LTD

FOR: INSTITUTIONAL CLIENT ENTERPRISE

Authorized Executive Representative
Executive Treasury & Compliance Committee